

items / objects: 1 2 3 4 | max weight = 5
 weight: 3 2 5 4 | n = 4
 value / profit: 4 3 6 5

		0	1	2	3	4	5
0		0	0	0	0	0	0
1		0	0	0	4	4	4
2		0	0	3	4	4	7
3		0	0	3	4	4	7
4		0					

But since the past value/ profit is greater we take the max

Here, it was supposed to be 6

items / objects:

1

2

3

4

max weight = 5

Weight:

3

2

5

4

$n = 4$

Value / Profit:

4

3

6

5

		0	1	2	3	4	5
0	0	0	0	0	0	0	0
1	0	0	0	0	4	4	4
2	0	0	3	4	4	7	
3	0	0					
4	0						

$(3+4)$

$5-2=3$

3128

items / objects: 1 2 3 4

Weight: 3 2 5 4

Value / Profit: 4 3 6 5

max weight = 5

$n = 4$

1st
15
20

31.8

		0	1	2	3	4	5
0	0	0	0	0	0	0	0
1	0	0	0	4	4	4	4
2	0	0	3	4	4	7	7
3	0	0	3	4	4	7	7
4	0	0	3	4	5	7	7

Max value we can take

Weight: 3 2 5 1
Value/Profit: 4 3 6 5

Made with KINEMASTER

$i \downarrow w \rightarrow$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	<u>7</u>

$$B[i, w] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$B[4, 4] = \max(B[3, 4], B[3, 4 - 4] + 5)$$

$$= \max(4, B[3, 0] + 5)$$

$$= \max(4, 0 + 5 = 5)$$

$$= \max(4, 5)$$

Value/ Profit: 4 3 6 5

0	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	7

Backtracking

if we take the item, we set it 1
otherwise we set is 0

$$B[i, w] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$B[4, 4] = \max(B[3, 4], B[3, 4 - 4] + 5) \quad | \quad B[4, 5] = \max(B[3, 5], B[3, 5 - 4] + 5)$$

$$= \max(4, B[3, 0] + 5) \quad | \quad = \max(B[3, 5], B[3, 1] + 5)$$

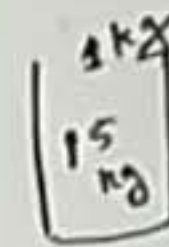
$$= \max(4, 5) \quad | \quad = \max(7, B[3, 0] + 5)$$

$$= \max(4, 5) \quad | \quad = \max(7, 5)$$

$$= 5 \quad | \quad = 7$$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

$n = 4$



Made with KINEMASTER

$i \backslash w$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	7

Since the previous value is equal to current value we do not take the corresponding item of current value

item: 1 2 3 4
0

$$B[i, w] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$\begin{aligned}
 B[4, 4] &= \max(B[3, 4], B[3, 4-4] + 5) \\
 &= \max(4, B[3, 0] + 5) \\
 &= \max(4, 0 + 5 = 5) \\
 &= \max(4, 5) \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 B[4, 5] &= \max(B[3, 5], B[3, 5-4] + 5) \\
 &= \max(B[3, 5], B[3, 1] + 5) \\
 &= \max(7, 0 + 5 = 5) \\
 &= \max(7, 5) \\
 &= 7
 \end{aligned}$$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

$n = 4$

31x2
15
10

Made with KINEMASTER

$i \downarrow w \rightarrow$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	7

item: 1 2 3 4
0 0

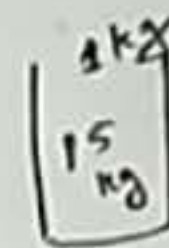
$$B[i, w] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$\begin{aligned}
 B[4, 4] &= \max(B[3, 4], B[3, 4-4] + 5) \\
 &= \max(4, B[3, 0] + 5) \\
 &= \max(4, 0 + 5) \\
 &= \max(4, 5) \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 B[4, 5] &= \max(B[3, 5], B[3, 5-4] + 5) \\
 &= \max(B[3, 5], B[3, 1] + 5) \\
 &= \max(7, 0 + 5) \\
 &= \max(7, 5)
 \end{aligned}$$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

$n = 4$



Made with KINEMASTER

$i \setminus w$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
✓ 2	0	0	3	4	4	7
✗ 3	0	0	3	4	4	7
✗ 4	0	0	3	4	5	7

Since the previous value is different, we can take the current value's corresponding item

item: 1 2 3 4
1 0 0

$$B[i, w] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$B[4, 4] = \max(B[3, 4], B[3, 4-4] + 5)$$

$$= \max(4, B[3, 0] + 5)$$

$$= \max(4, 0 + 5 = 5)$$

$$B[4, 5] = \max(B[3, 5], B[3, 5-4] + 5)$$

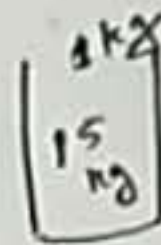
$$= \max(B[3, 5], B[3, 1] + 5)$$

$$= \max(7, 0 + 5 = 5)$$

$$= \max(7, 5)$$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

$n = 4$



Made with KINEMASTER

item: 1 2 3 4
1 0 0

		0	1	2	3	4	5
0	0	0	0	0	0	0	0
1	0	0	0	4	4	4	
2	0	0	3	4	4	7	
3	0	0	3	4	4	7	
4	0	0	3	4	5	7	

$$7 - 3 = 4$$

$$B[i] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$= \max(B[3, 4], B[3, 4 - 4] + 5) \quad | \quad B[4, 5] = \max(B[3, 5], B[3, 5 - 4] + 5)$$

$$= \max(4, B[3, 0] + 5)$$

$$= \max(4, 0 + 5 = 5)$$

$$= \max(4, 5)$$

$$= 5$$

$$= \max(B[3, 5], B[3, 1] + 5)$$

$$= \max(7, 0 + 5 = 5)$$

$$= \max(7, 5)$$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

15
10

Made with KINEMASTER

0	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	7

item: 1 2 3 4
1 0 0

$7 - 3 = 4$

We look for this value in the previous value's row, and set the pointer to that value

$= \max (\dots)$
 $= \max (\dots)$
 $= \max (\dots)$
 $= \max (\dots)$
 $= \max (\dots)$
 $= \max (\dots)$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

$n = 4$

15
10

Made with KINEMASTER

$i \backslash w$	0	1	2	3	4	5
0	0	0	0	0	0	0
✓ 1	0	0	0	4	4	4
✓ 2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	7

item: 1 2 3 4
1 1 0 0

$$7 - 3 = 4$$

$$B[i, w] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$B[4, 5] = \max(B[3, 5], B[3, 5 - 4] + 5)$$

$$= \max(B[3, 5], B[3, 1] + 5)$$

$$= \max(7, 0 + 5 = 5)$$

$$= \max(7, 5)$$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

$n = 4$

Made with KINEMASTER

item: 1 2 3 4
1 1 0 0

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	0	4	4	4
2	0	0	3	4	4	7
3	0	0	3	4	4	7
4	0	0	3	4	5	7

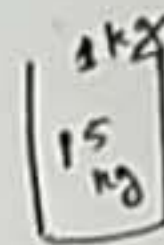
$7 - 3 = 4$
 $4 - 4 = 0$

$$\begin{aligned}
 &= \max(B[i-1, w], B[i-1, w - w[i]] + v[i]) \\
 &= \max(B[3, 4], B[3, 4 - 4] + 5) \quad | \quad B[4, 5] = \max(B[3, 5], B[3, 5 - 4] + 5) \\
 &= \max(4, B[3, 0] + 5) \\
 &= \max(4, 0 + 5 = 5) \\
 &= \max(4, 5) \\
 &= 5
 \end{aligned}$$

$$\begin{aligned}
 &= \max(B[3, 5], B[3, 1] + 5) \\
 &= \max(7, 0 + 5 = 5) \\
 &= \max(7, 5)
 \end{aligned}$$

Weight: 3 2 5 4
Value/Profit: 4 3 6 5

$n = 4$



Made with KINEMASTER

		0	1	2	3	4	5
0	0	0	0	0	0	0	0
1	0	0	0	4	4	4	4
2	0	0	3	4	4	7	7
3	0	0	3	4	4	7	7
4	0	0	3	4	5	7	7

item: 1 2 3 4
1 1 0 0

$$7 - 3 = 4$$

$$4 - 4 = 0$$

$$= \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$$

$$B[4, 5] = \max(B[3, 4], B[3, 4 - 4] + 5)$$

$$= \max(4, B[3, 0] + 5)$$

$$= \max(4, 0 + 5 = 5)$$

$$= \max(4, 5)$$

$$= 5$$

$$B[4, 5] = \max(B[3, 5], B[3, 5 - 4] + 5)$$

$$= \max(B[3, 5], B[3, 1] + 5)$$

$$= \max(7, 0 + 5 = 5)$$

$$= \max(7, 5)$$

Value / Profit: 4 3 6 5 | $n=4$

15 kg

Made with KINEMASTER

$i \backslash w$	0	1	2	3	4	5
0	0	0	0	0	0	0
✓ 1	0	0	0	4	4	4
✓ 2	0	0	3	4	4	7
✗ 3	0	0	3	4	4	7
✗ 4	0	0	3	4	5	7

item: 1 2 3 4

1	1	0	0
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$4 + 3 = 7$

$7 - 3 = 4$
 $4 - 4 = 0$

$B[i, w] = \max(B[i-1, w], B[i-1, w - w[i]] + v[i])$

$B[4, 4] = \max(B[3, 4], B[3, 5 - 4] + 5)$
 $= \max(4, B[3, 0] + 5)$
 $= \max(4, 0 + 5 = 5)$
 $= \max(4, 5)$